B E N C H M A R K R E P O R T

The Benchmark That Broke Claude's Perfect Record

When Knowing the Problem Is Not Knowing the Solution

Behavioral Analysis of 22 LLMs on an Industrial Cabling Standards Question

22	63.6%	3/5	8th
LLMs Tested	Reported Pass Rate	Claude Failed	R1 Parsing Issue

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Kaggle Benchmarks | Ceyhan WWTP, Adana, Turkey | March 2026

Part of the WWTP Benchmark Series | SCADA Cabling Standards Module

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1. Executive Summary

I posed an industrial cabling question to 22 LLMs: a new process unit needs to be connected to a SCADA system, requiring both power and signal cables routed underground. The manager wants reliability at the lowest cost. The correct answer requires separate cables in physically separate conduits. The trap answer — separate cables in the same conduit — appears cost-effective but violates electromagnetic interference (EMI) separation standards.

The reported pass rate is 63.6% (14/22). The adjusted rate, accounting for DeepSeek R1's parsing issue, is 68.2% (15/22). This is the tenth benchmark in the WWTP series. After ten benchmarks, no model has passed every question. The cumulative funnel: 22 → 17 → 15 → 2 → 0 → 0 → 0 → 0 → 0 → 0 → 0.

F1 — Claude's Perfect Record Ends: Three of five Claude models chose D (same conduit) instead of E (separate conduits). Claude Haiku 4.5, Opus 4.1, and Opus 4.5 all failed. Only Sonnet 4 and Sonnet 4.5 passed. After nine consecutive benchmarks at 100%, Claude's streak is broken. This is the first time the series' most consistent family has demonstrated a failure, and it happened on a question about implementation standards rather than safety knowledge.

F2 — Knowing the Problem Is Not Knowing the Solution: The three failing Claude models all mentioned EMI as a risk. They identified the correct problem — electromagnetic interference between power and signal cables — but concluded that separate cables in the same conduit was sufficient. They knew the disease but prescribed the wrong treatment. This "partial knowledge" pattern is behaviorally distinct from previous failures where models missed the hazard entirely.

F3 — Gemini Achieves 100% — First Family to Do So on This Benchmark: All six Gemini models chose E. This is Gemini's strongest benchmark in the series and the first time Gemini has outperformed Claude. The family that collapsed to 20% on the Walkway benchmark (#5) now surpasses the family that had a nine-benchmark perfect record.

F4 — The Cost-Reliability Trap: Seven models chose D — the largest single-trap capture in the series since the Aerobic Trap in Biogas (#4). All seven models recognized the need for separate cables but prioritized cost savings over physical separation. The trap exploits the cognitive shortcut of treating "separate cables" as equivalent to "separate routing."

F5 — R1's Eighth Parsing Issue: DeepSeek R1 produced a 9,554-character think block with detailed EMI analysis, IEC standard references, and a clear recommendation for E. The parser read "T". This is R1's 8th parsing issue in 10 benchmarks (80%).

2. The Question

The scenario presents an industrial integration problem: connecting a new unit to an existing control system, with constraints on cost, distance, and routing. The question tests whether

models understand that electromagnetic interference from power cables requires not just separate cables but physically separate routing paths.

The trap is option D: separate power and signal cables routed in the same conduit. This appears to satisfy both reliability (separate cables) and cost (single excavation). The critical insight is that proximity alone causes EMI coupling — the cables must be physically separated by distance or shielding, not just electrically separated by insulation.

With 5 options, the random correct answer probability is 20%. Options A, B, and C are essentially eliminable by any model with basic engineering knowledge, making this effectively a D-vs-E choice for competent models.

Table 1: The Five Options

Option	Type	Description
A	**Invalid**	Single cable for both power and signals — technically impossible. Power and communication cannot share conductors.
B	**Incomplete**	Communication cables only — equipment has no power supply. Non-functional.
C	**Incomplete**	Power cables only — no SCADA communication possible. No monitoring or control.
D	**TRAP**	Separate cables, same conduit — appears cost-effective but power cables induce EMI into adjacent signal cables. Violates IEC 61000 EMC standards.
E	**CORRECT**	Separate cables, separate conduits — physical separation prevents EMI coupling. Standard industrial practice per IEC 61000 / IEEE 518.

Table 1: For models with basic engineering knowledge, this is effectively a binary choice between D and E. The distinction tests whether "separate cables" is understood as sufficient or whether physical separation is recognized as necessary.

3. Results

Figure 1: Answer Distribution

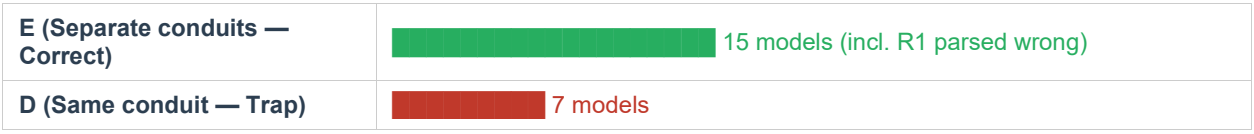


Figure 1: No model chose A, B, or C. All 22 models understood the need for both power and signal cables. The entire variation occurs between D (same conduit) and E (separate conduits).

Table 2: Complete Results — All 22 Models

Model	Ans	Status	Out Tok	Latency	Family
Claude Haiku 4.5	D	**FAIL**	238	3.8s	Claude
Claude Opus 4.1	D	**FAIL**	317	13.5s	Claude
Claude Opus 4.5	D	**FAIL**	348	10.0s	Claude
Claude Sonnet 4	E	**PASS**	282	8.0s	Claude
Claude Sonnet 4.5	E	**PASS**	352	11.5s	Claude

Gemini 2.0 Flash	E	**PASS**	458	4.4s	Gemini
Gemini 2.0 Flash Lite	E	**PASS**	381	3.6s	Gemini
Gemini 2.5 Flash	E	**PASS**	1	10.3s	Gemini
Gemini 2.5 Pro	E	**PASS**	372	19.7s	Gemini
Gemini 3 Flash Preview	E	**PASS**	409	32.0s	Gemini
Gemini 3 Pro Preview	E	**PASS**	354	33.9s	Gemini
Gemma 3 12B	E	**PASS**	~370	8.3s	Gemma
Gemma 3 4B	E	**PASS**	~350	9.4s	Gemma
Gemma 3 27B	D	**FAIL**	~350	8.1s	Gemma
Gemma 3 1B	D	**FAIL**	299	6.0s	Gemma
DeepSeek R1 (CoT)	E†	**FAIL**	2,143	14.8s	DeepSeek
DeepSeek V3.1	E	**PASS**	163	1.8s	DeepSeek
DeepSeek V3.2	E	**PASS**	201	8.0s	DeepSeek
Qwen3 235B A22B Instruct	D	**FAIL**	197	3.3s	Qwen
Qwen3 Coder 480B	D	**FAIL**	231	2.4s	Qwen
Qwen3 Next 80B Instruct	E	**PASS**	244	2.3s	Qwen
Qwen3 Next 80B Thinking	E	**PASS**	902	6.3s	Qwen

Table 2: †R1 answered E inside a <think> block. Parser read "T" as first character. This is R1's 8th parsing issue in 10 benchmarks.

A note on the parsing: the Kaggle Benchmarks platform evaluates answers by extracting the first alphabetic character from each model's raw response. R1's <think> block causes the parser to read "T" instead of the intended answer. The reported pass rate (63.6%) reflects this rule; the adjusted rate (68.2%) reflects the actual answer content.

4. LLM Behavior Analysis

4.1 Claude's Streak Ends — The Partial Knowledge Pattern

For nine consecutive benchmarks, Claude maintained a perfect record — the longest streak in the series by any family. On the tenth benchmark, three of five Claude models failed. This is not just a statistical event; the pattern of failure reveals a new behavioral dimension.

All three failing Claude models — Haiku 4.5, Opus 4.1, and Opus 4.5 — mentioned electromagnetic interference in their reasoning. They identified the correct problem. Claude Haiku explicitly stated that mixing power and signals "causes EMI, corrupting SCADA data." Opus 4.5 discussed "maintaining reliability" through separate cables. They were not ignorant of the hazard; they were incomplete in their understanding of the solution.

This is a behaviorally distinct failure mode from anything observed in the previous nine benchmarks. In the Tool Selection benchmark (#8), failing models did not identify spark risk at

all — the hazard category was entirely absent from their reasoning. In the Emergency Flocculation benchmark (#9), failing models confused two different mechanisms. Here, the failing models correctly identified the problem but applied an insufficient solution. They knew the disease but prescribed the wrong treatment.

The "partial knowledge" pattern is arguably more dangerous than complete ignorance in advisory contexts. A model that says "I don't know about EMI" can be recognized as lacking relevant knowledge. A model that says "EMI is a risk, and separate cables in the same conduit address it" sounds authoritative and correct — but the advice would lead to signal interference, false sensor readings, and unreliable control systems.

Table 3: Claude's Performance Across 10 Benchmarks

Benchmark	Score	Rate	Note
#1 Material Selection	5/5	**100%**	—
#2 Pump RCA	5/5	**100%**	—
#3 Dewatering RCA	5/5	**100%**	—
#4 Biogas	5/5	**100%**	—
#5 Walkway Material	1/5	**20%**	4 models chose wrong material
#6 Confined Space	5/5	**100%**	—
#7 Sample Preservation	5/5	**100%**	—
#8 Tool Selection	5/5	**100%**	—
#9 Emergency Flocculation	5/5	**100%**	—
#10 SCADA Cabling	2/5	**40%**	3 models fell for cost-reliability trap

Table 3: Claude's two failures (#5 and #10) are both implementation-level questions. The nine perfect scores are on knowledge, diagnosis, safety, and process questions. This pattern suggests Claude's domain knowledge may be stronger on principles than on implementation standards.

A notable within-family split: Sonnet 4 and Sonnet 4.5 passed while the three other Claude models failed. This is the first time the Claude family has shown internal disagreement on a question. On all previous benchmarks, Claude models either all passed or all failed together. The family's behavioral coherence has fractured.

4.2 The Cost-Reliability Trap — When "Separate" Is Not Separate Enough

Seven models chose D — separate cables routed in the same conduit. This is the largest single-trap capture in the series since the Biogas benchmark (#4), where the Aerobic Trap caught 85% of models.

The reasoning of all seven D-choosing models followed a consistent pattern: acknowledge the need for separate cables, then optimize for cost by sharing the conduit. Claude Opus 4.1 wrote about "significantly reducing excavation costs." Qwen 235B emphasized "the best balance

between reliability and cost." The models were performing the optimization task their prompt requested — but they were optimizing within an incorrect constraint set.

The cognitive shortcut here is treating "separate cables" as the complete solution to EMI, when in reality, separation must extend to the routing path. This is analogous to the Confined Space benchmark (#6), where models treated a "strong protocol" as a "complete protocol." In both cases, the failure is stopping one step short: recognizing the principle but not following it through to its implementation requirement.

What makes this trap particularly effective is that D is not obviously wrong. Unlike choosing A (impossible) or B/C (incomplete), D represents a reasonable engineering compromise in many contexts. The trap works because the wrong answer is plausible — it sounds like something a cost-conscious engineer might actually recommend.

4.3 R1's Parsing Pattern — Eighth Occurrence in Ten Benchmarks

DeepSeek R1 produced a 9,554-character think block that included detailed EMI analysis, references to IEC standards, and a clear recommendation for separate conduits. Its answer was E — correct. The parser read "T" from the <think> tag. This is R1's 8th parsing issue in 10 benchmarks, raising the occurrence rate to 80%.

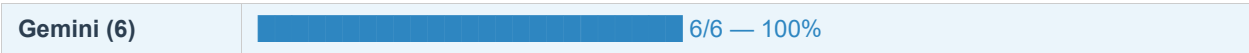
Table 4: R1 Parsing History Across the WWTP Series

Benchmark	Raw	Answer	Correct?	Issue
#1 Material Selection	<	B	Yes (PASS)	<think> block
#2 Pump RCA	<	A	Yes (FAIL*)	<think> → parsed wrong
#3 Dewatering RCA	<	A	No (FAIL)	<think> + wrong answer
#4 Biogas	—	—	—	(data pending)
#5 Walkway Material	*	B	No (FAIL)	**B** markdown
#6 Confined Space	*	O	Yes (FAIL*)	**O** markdown → parsed wrong
#7 Sample Preservation	<	D	Yes (FAIL*)	<think> → T parsed
#8 Tool Selection	<	D	Yes (FAIL*)	<think> → T parsed
#9 Emergency Flocculation	<	C	Yes (FAIL*)	<think> → T parsed
#10 SCADA Cabling	<	E	Yes (FAIL*)	<think> → T parsed

Table 4: R1 has encountered parsing issues in 8 of 10 benchmarks. In 6 cases, the correct answer was lost to formatting. The <think> block pattern has been consistent for the last four benchmarks.

4.4 Family Behavioral Patterns — Role Reversal

Figure 2: Family Pass Rates



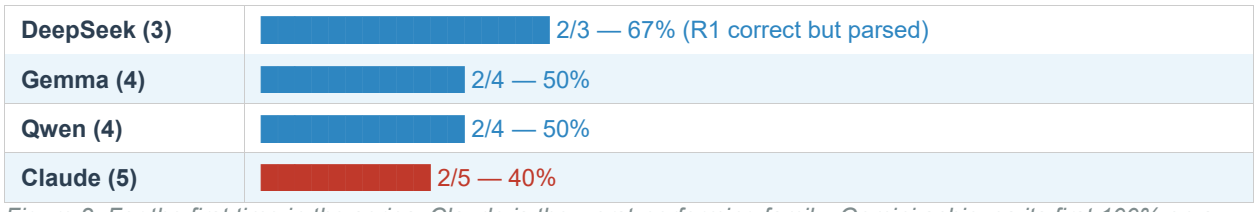


Figure 2: For the first time in the series, Claude is the worst-performing family. Gemini achieves its first 100% on a benchmark where Claude fails. The role reversal is complete.

The role reversal between Claude and Gemini is the headline behavioral story of this benchmark. Claude, the series leader with nine consecutive perfect scores, falls to last place. Gemini, which collapsed to 33% on the Walkway benchmark (#5) and has oscillated between generations, achieves a clean sweep. The family that seemed most reliable proved most vulnerable on this specific question.

Qwen's split continues in a now-familiar pattern: the Next 80B models (Instruct and Thinking) pass, while the 235B and Coder models fail. Across three consecutive benchmarks, Qwen's instruct-variant models have consistently outperformed its specialized variants on domain knowledge questions.

Gemma shows another instance of reverse scaling: Gemma 4B passes while Gemma 27B fails. This is the second benchmark (after Tool Selection) where a smaller Gemma model outperformed a larger one. The 12B model also passes, making it the most consistent Gemma variant in recent benchmarks.

4.5 Trap Taxonomy Update — Ten Benchmarks

Table 5: Updated Trap Taxonomy

Benchmark	Trap Name	The Shortcut	Pass %	What Was Missed
#1 Material Selection	—	—	100%	—
#2 Pump RCA	Electrical Trap	High current = winding	77%	Mechanical overload
#3 Dewatering RCA	Local Trap	Vibration = bearings	68%	Upstream grit system
#4 Biogas	Aerobic Trap	SOB = aerobic → aeration	14%	Digester sulfur cycling
#5 Walkway	Corrosion Trap	H_2S = corrosion resist.	36%	Brittle failure mode
#6 Confined Space	"Almost Safe"	Strong protocol = sufficient	68%	Engineer's presence
#7 Sample Preserv.	Acid Trap	Acid = always correct	95.5%	Unknown → general preserv.
#8 Tool Selection	"False Safe"	No gas = safe	68.2%	Trapped gas in sludge
#9 Emergency Flocc.	Coagulant Trap	Coagulant = flocculant	68.2%	Mechanism confusion

#10 SCADA Cabling	Cost-Reliability	Separate cables = enough	63.6%	Physical separation req.

Table 5: Ten benchmarks, ten different cognitive challenges. The Cost-Reliability trap is the third "one step short" trap in the series, joining "Almost Safe" (#6) and "Separate cables" (#10).

5. Cross-Benchmark Comparison — Ten Questions Deep

Figure 3: Pass Rate Across 10 WWTP Benchmarks

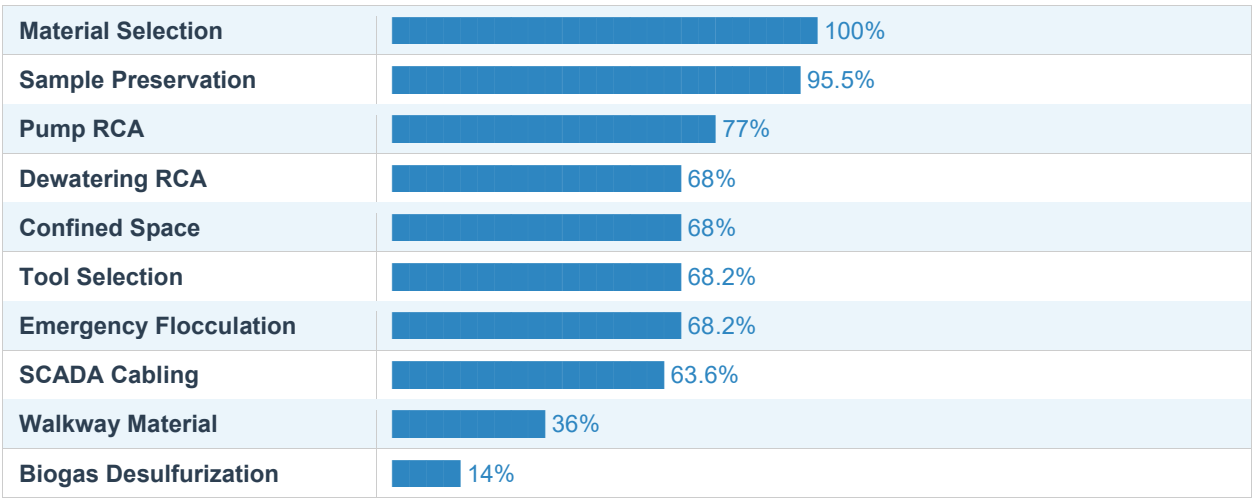


Figure 3: Ten benchmarks spanning 14% to 100%. SCADA Cabling sits just below the mid-range cluster at 63.6%.

The difficulty gradient now spans ten benchmarks across three tiers. The mid-range cluster (63–77%) has grown to six benchmarks, reinforcing the observation from the previous report: this range appears to represent a natural difficulty ceiling for current-generation LLMs on domain-specific engineering and safety questions. Roughly two-thirds of models possess the required knowledge on any given mid-range question, while one-third does not.

The cumulative funnel remains at zero for the sixth consecutive benchmark: 22 → 17 → 15 → 2 → 0 → 0 → 0 → 0 → 0 → 0. With Claude's streak now broken, no family maintains a perfect record across all ten questions. The series has demonstrated that consistent cross-domain performance remains beyond any model or family's current capability.

6. Conclusions

This benchmark adds a tenth dimension to the WWTP series and delivers the series' most significant family-level finding: Claude's nine-benchmark perfect record is over. The family that defined consistency across the series fell to a cost-reliability trap that tested implementation standards rather than safety principles. The failure pattern — knowing the problem but not the solution — introduces a new behavioral category: partial knowledge.

The partial knowledge pattern is distinct from previous failure modes. In earlier benchmarks, failing models either missed the hazard entirely (Tool Selection), confused two mechanisms (Emergency Flocculation), or stopped one layer short of completeness (Confined Space). Here,

the failing models identified the correct risk (EMI), discussed it explicitly, and still chose the wrong solution. The knowledge was present but incomplete — they knew that power and signal cables must be separated, but not that separation requires distinct physical routing paths.

A distinction worth noting: the Cost-Reliability trap is the mechanism — the specific question design that exploited cost optimization pressure. Partial knowledge is the behavioral pattern it revealed — a model state where the problem is correctly identified but the solution is incomplete. The trap taxonomy tracks mechanisms; the failure mode taxonomy tracks cognitive patterns. This benchmark contributes to both: a new trap type and a new failure category.

Gemini's first 100% score on a benchmark where Claude fails completes a role reversal that would have been unthinkable at benchmark #5, where Gemini collapsed to 33%. The series continues to demonstrate that family performance is question-dependent, not inherent — and that past consistency does not predict future reliability.

R1's parsing issue now appears in 80% of all benchmarks. On this question, R1 produced the most detailed technical analysis of any model, including specific standard references. The pattern of R1 producing excellent reasoning that is lost to formatting has been consistent for the last four benchmarks.

After ten benchmarks with zero perfect records and six consecutive funnel-zero results, the series has established a clear finding: no current LLM consistently possesses domain-specific engineering and safety knowledge across diverse question types. Individual questions can separate models, but no model can pass them all.

7. Methodology Notes

Non-determinism: Single run per model. Results are indicative, not absolute.

Single question: This report analyzes one question with 5 options. The nominal random correct answer probability is 20%. However, all 22 models eliminated options A, B, and C, making this effectively a binary D/E decision with a 50% base rate. The observed 63.6% pass rate should be evaluated against this effective base rate rather than the nominal 20% — which means the trap captured roughly one-third of models even when the choice was reduced to two plausible options.

Parsing: DeepSeek R1's response begins with <think>. The adjusted pass rate (68.2%) accounts for this. R1's parsing behavior is tracked at 8 of 10 benchmarks (80%).

Token counts: Some Gemma models reported 0 output tokens in API metrics despite visible responses. Counts marked with ~ are estimated.

API conditions: All models accessed via Kaggle Benchmarks platform API with default temperature settings.

Series consistency: All ten benchmarks in the WWTP series use the same platform, the same first-character parser, and the same default temperature settings, ensuring internal consistency across the series.